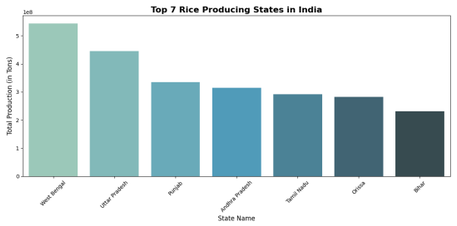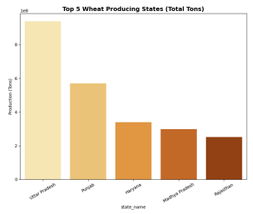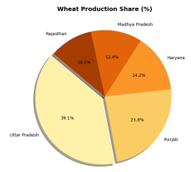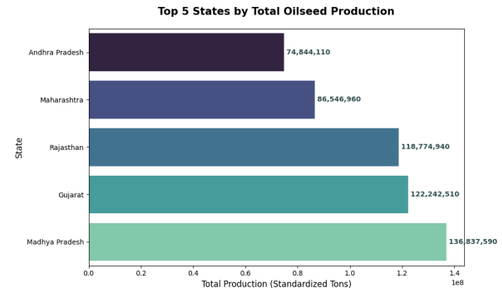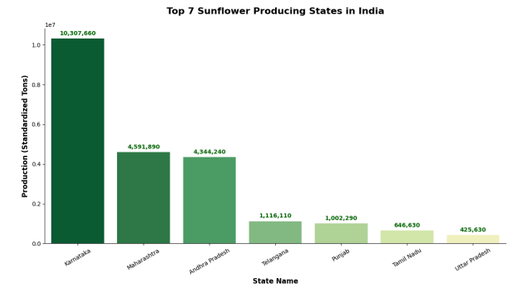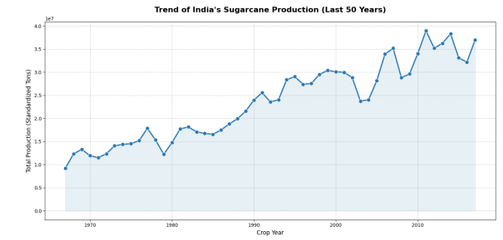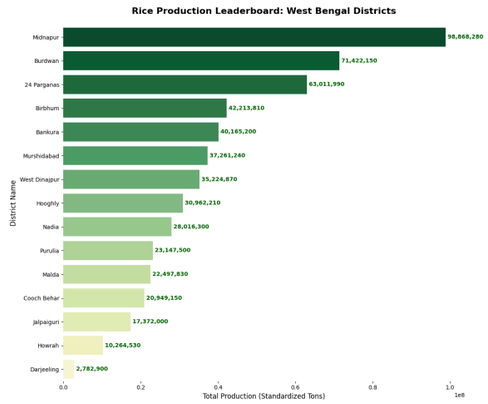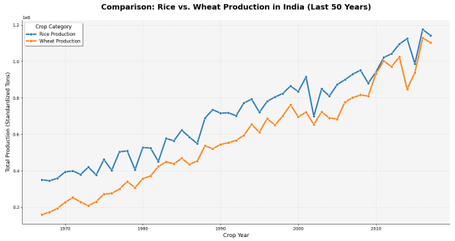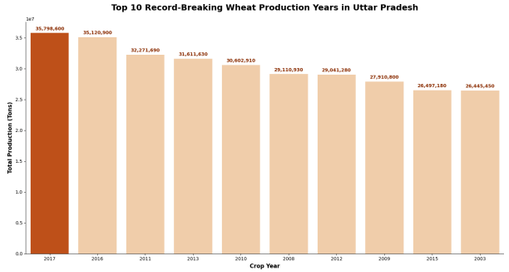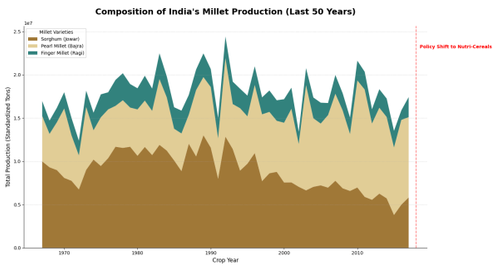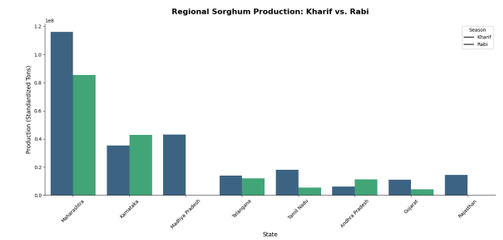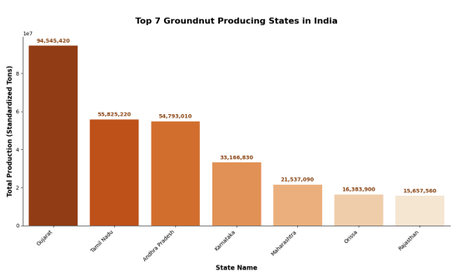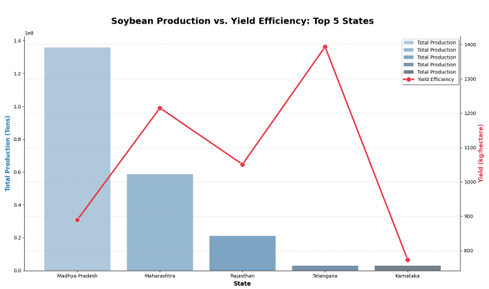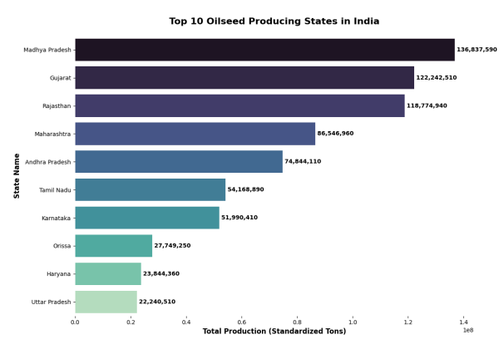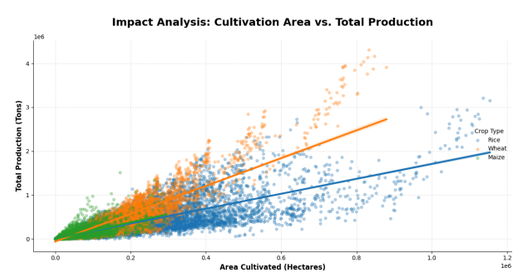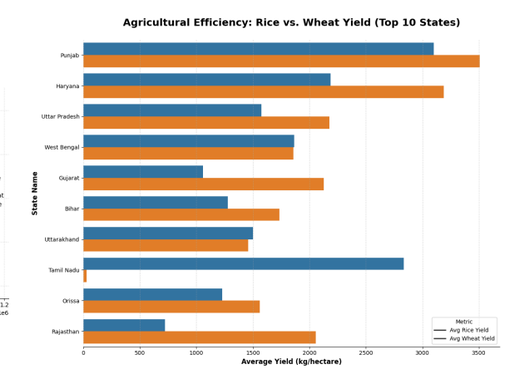AGRICULTURAL DATA ANALYSIS: 50-YEAR INSIGHTS